

CS410 Composite

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I. QUESTION (MIN: 4 MAX: 20)

Let $A_5 = \{a^p b^q c^r \mid \text{where } p + q + r \text{ is a multiple of } 10\}$. Draw a DFA that recognizes A_5 .

II. QUESTION (MIN 3 MAX: 15)

Let $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Let $B_{(8,3)} = \{x \mid x \text{ is a number in base 8 that is a multiple of } 3\}$. Draw an NFA that recognizes $B_{(8,3)}$.

III. QUESTION (MIN: 4 MAX: 20)

Give a CFG generating the language of strings with three times as many a's as b's. The alphabet is $\{a, b\}$.

IV. QUESTION (MIN: 3 MAX: 15)

Let $\Sigma = \{0, 1\}$. Give the PDA for the following language: $C = \{w \mid w \text{ contains at least three 0s and at most three 1s}\}$

V. QUESTION (MIN: 6 MAX: 30)

Describe a single tape TM for the following language: $E = \{w \# x \mid ww \text{ is a substring of } x \text{ for } w, x \in \{0, 1\}^*\}$ Draw a state diagram, give states, transitions, etc.